GlowAI

SKIN CONDITION CLASSIFICATION FROM SYNTHETIC DATA

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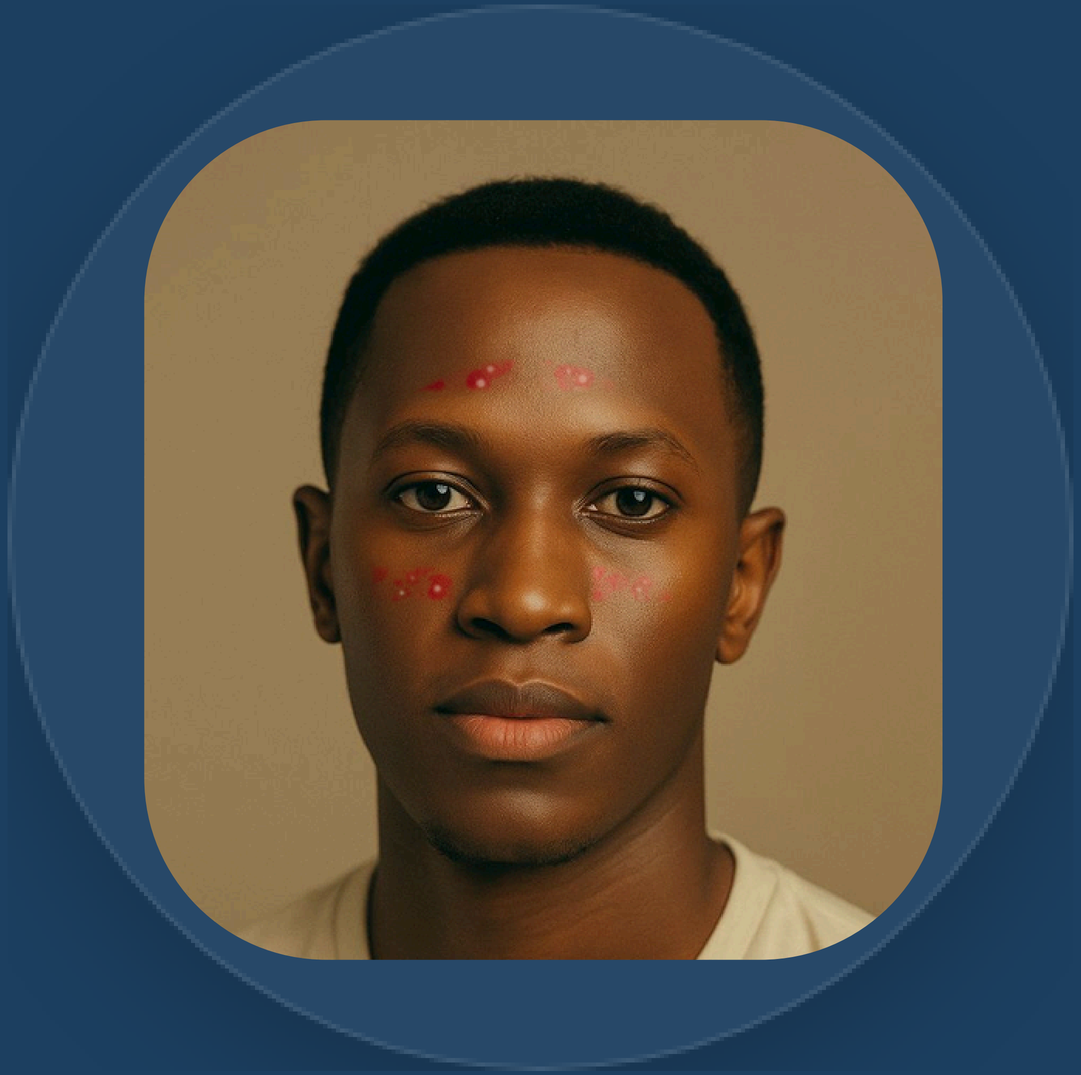
Project Definition

Goal & Motivation

- **Goal:** Classify face images into three classes – **acne, erythema, none**.
- **Motivation:** Quick pre-screening from consumer-grade images (selfies), not a medical diagnosis.

Input & Output

- **Input:** RGB face image (frontal selfie).
- **Output:** Predicted class + confidence score.



GlowAI Classifier

Acne - 72% confidence

Project Achievements & Novelty

Achievements

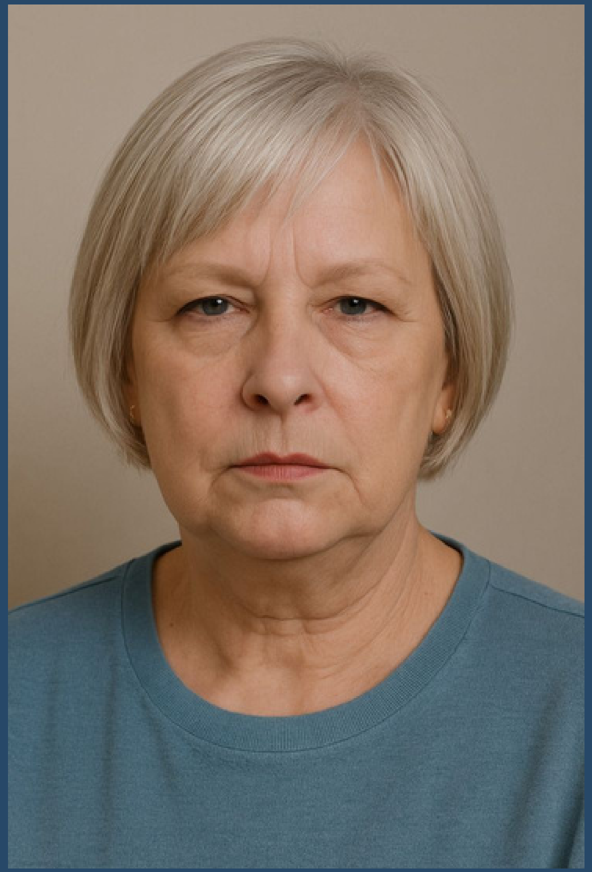
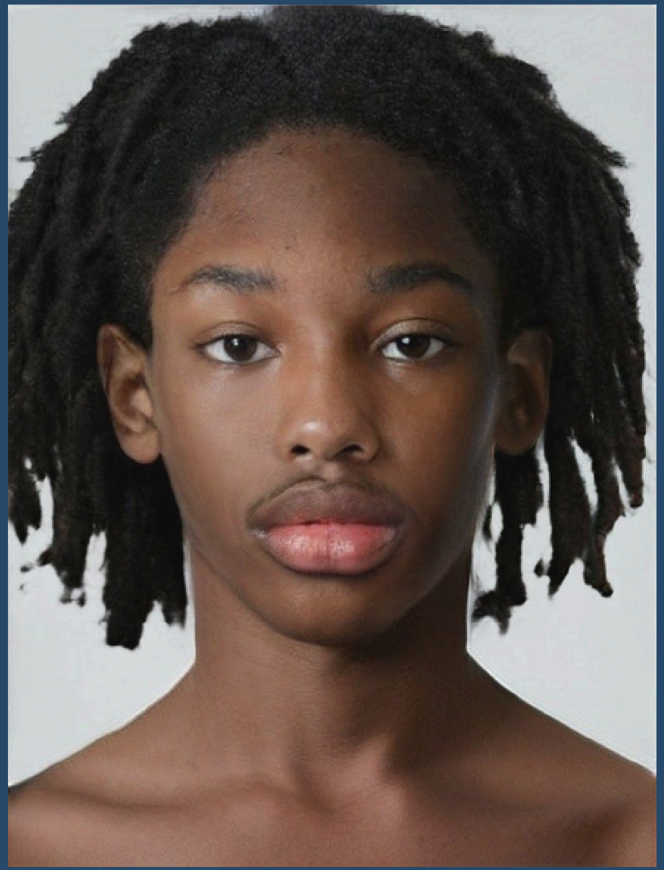
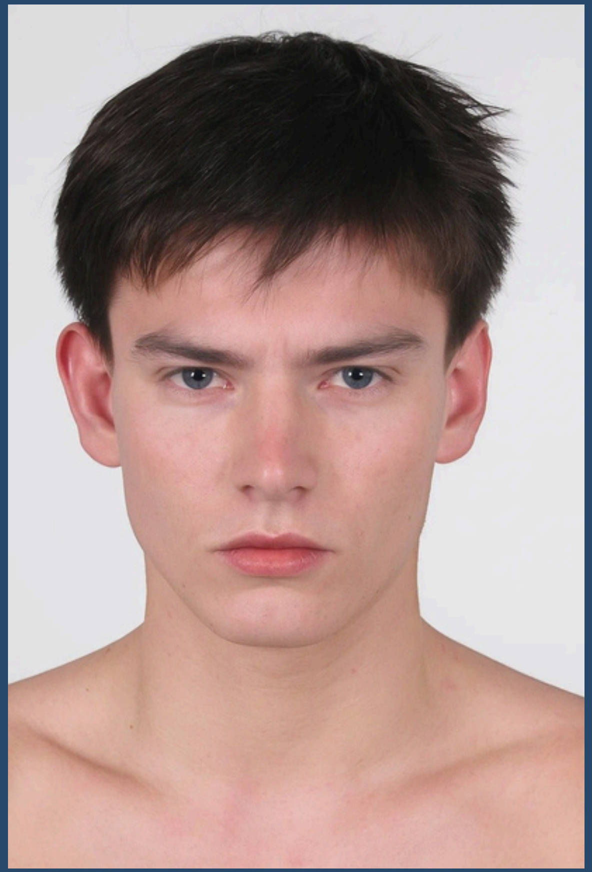
- Trained a 3-class classifier (EfficientNet-B2) combining real and synthetic data.
- Built an end-to-end pipeline: generation → training → evaluation → feedback.

Novelty

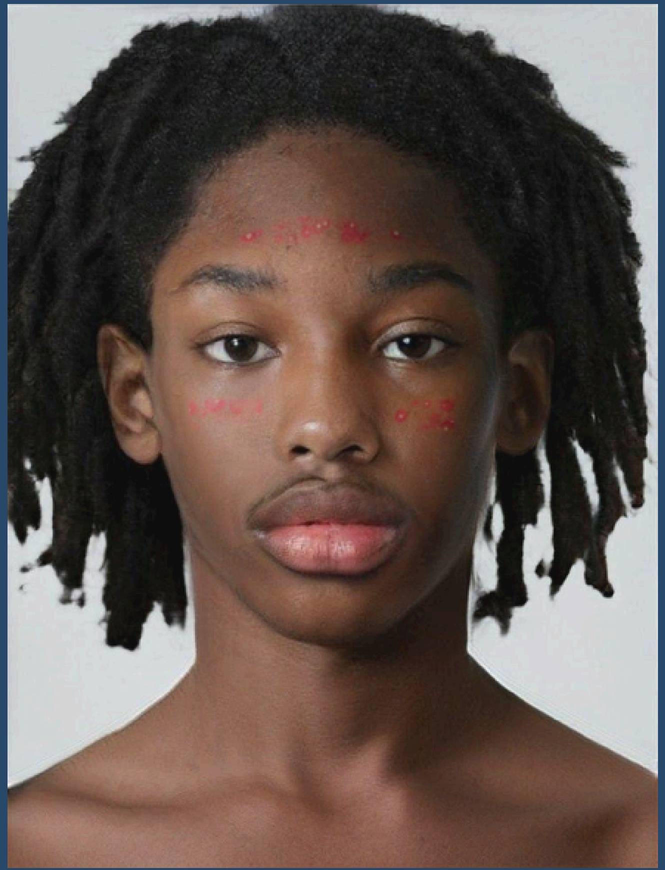
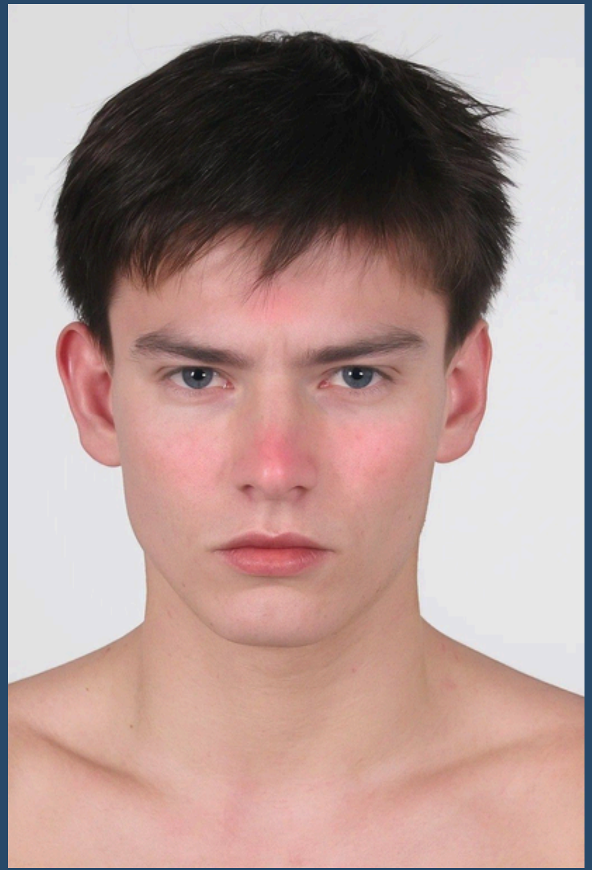
- **Custom Skin Simulator:** OpenCV-based tool for realistic acne & erythema control (location, spread, intensity).
- **Synthetic Data Engine:** Solving scarcity of labeled medical data.
- **Feedback Loop:** Manual set to guide improvements.

Comparison: Clean vs. Synthetic

BEFORE


AFTER

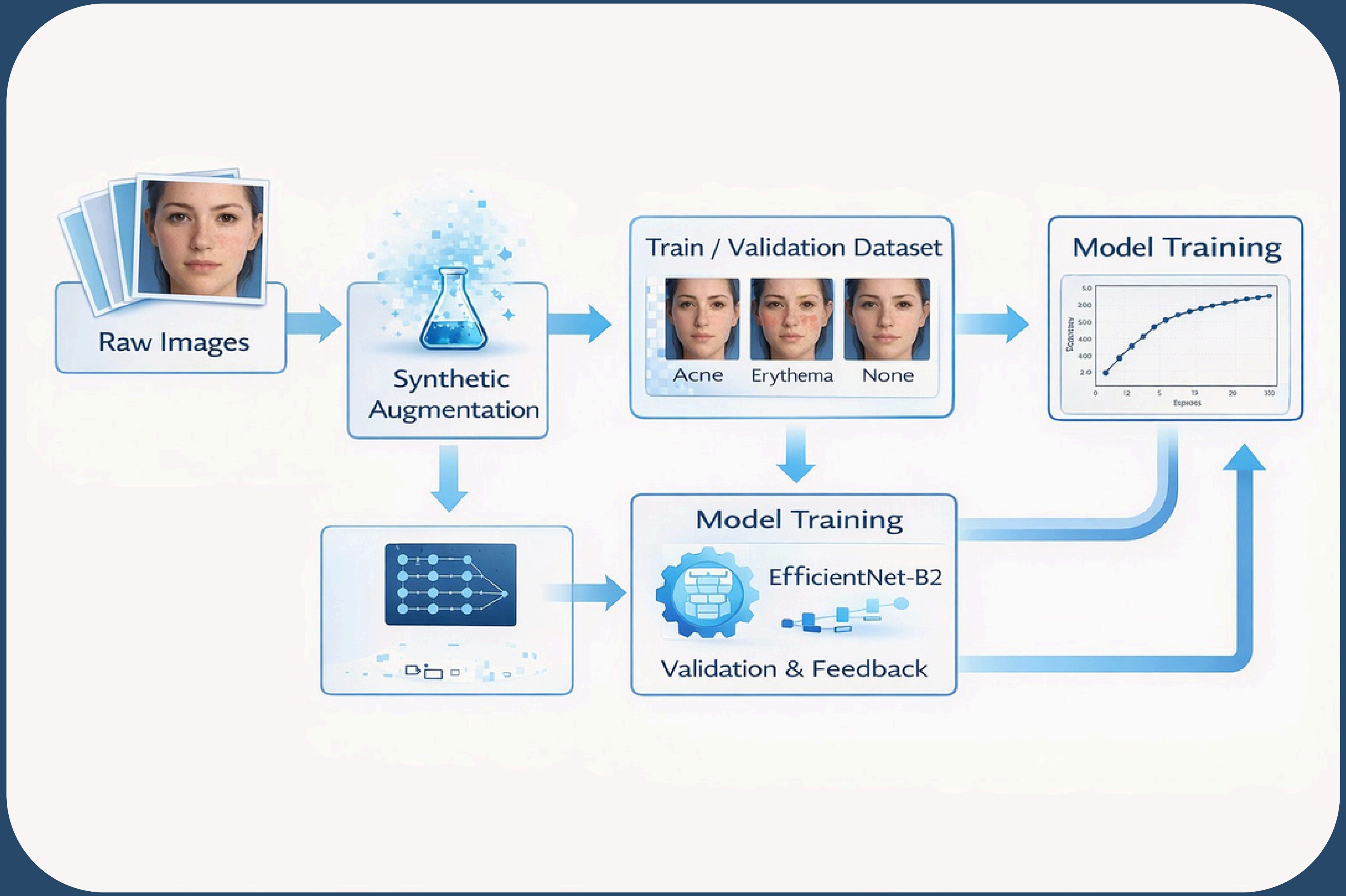

Methodology & Pipeline

Methodology

- **Data:** 3-class dataset (Acne/Erythema/None) mixed with synthetic data.
- **Preprocessing:** Resize 224×224, Normalize (ImageNet).
- **Augmentations:** Flip, rotation, color jitter (mimic selfies).
- **Training:** EfficientNet-B2, Two-stage fine-tuning, AdamW, Cross-Entropy.

Evaluation

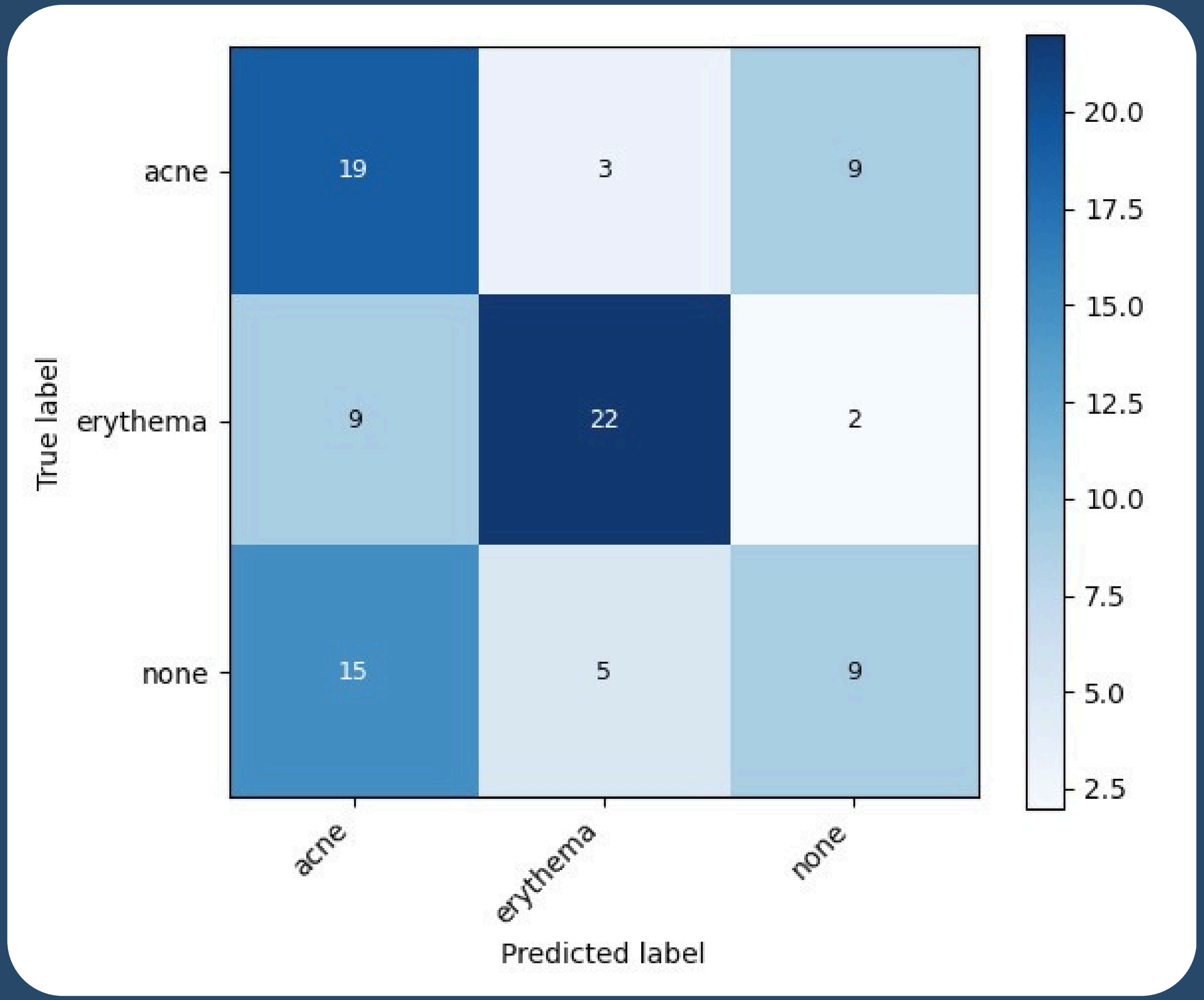
- Selected best model by validation accuracy.
- Analyzed per-class behavior via confusion matrix.



Results: Validation & Manual Test

Validation Results

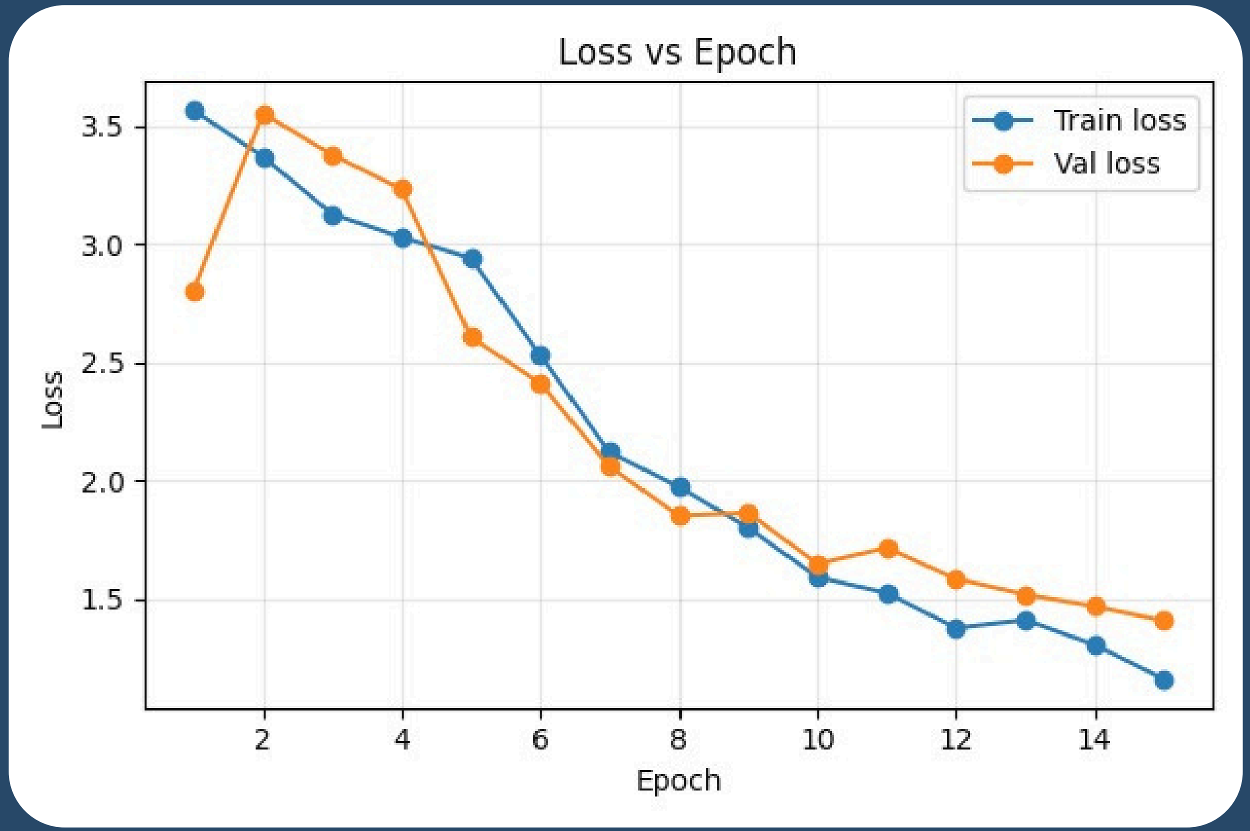
- Best Val Accuracy: **~54%**.
- Main issue: "None" often predicted as "Acne/Erythema".



Confusion Matrix

Manual Test (Real Images)

- **Acne:** 11/15 | **Erythema:** 8/9 | **None:** 17/17
- **Overall Accuracy: ~88%** on curated real-world set.
- Significantly better performance on clean input than raw validation metrics suggest.



Training vs Validation Accuracy

Conclusion & Future Work

🏁 Conclusion

- Built full synthetic+real classifier pipeline.
- Reached **~54% Val / ~88% Real Test** accuracy.
- Synthetic data is useful but highlights class overlap challenges.

⚠️ Limitations

- Separating mild conditions is difficult.
- Class imbalance impacts robustness.
- Single-label setup is a simplification of reality.

🚀 Future Work

- Expand expert-labeled datasets.
- Move to pixel-level segmentation.
- Use diffusion models for better realism.

"These improvements could lead to a more accurate and clinically relevant skin analysis system."